DS Assignment 4

Implement Berkeley algorithm for clock synchronization.

```
import java.util.*;

public class Berkeley {

    static int toSeconds(int h, int m, int s) {
        return h * 3600 + m * 60 + s;
    }

    static String toHMS(int total) {
        total = ((total % 86400) + 86400) % 86400;
        int h = total / 3600;
        int m = (total % 3600) / 60;
        int s = total % 60;
        return h + ":" + m + ":" + s;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Calendar cal = Calendar.getInstance();
        int masterTime = toSeconds(
                cal.get(Calendar.HOUR_OF_DAY),
                cal.get(Calendar.MINUTE),
                cal.get(Calendar.SECOND));

        System.out.println("Master time: " + toHMS(masterTime));
```

```
System.out.print("Enter number of nodes: ");
int n = sc.nextInt();

int[] nodeTime = new int[n];
int[] diff = new int[n];
int sum = 0;

for (int i = 0; i < n; i++) {
    System.out.println("Enter time for Node " + (i + 1) + " (h m s): ");
    nodeTime[i] = toSeconds(sc.nextInt(), sc.nextInt(), sc.nextInt());
}

System.out.println("\n--- Time Differences ---");

for (int i = 0; i < n; i++) {
    diff[i] = nodeTime[i] - masterTime;
    sum += diff[i];
    System.out.println("Node " + (i + 1) + " difference: " + diff[i] + "s");
}

int avg = sum / (n + 1);
System.out.println("\nAverage correction: " + avg + "s");

System.out.println("\n--- Corrections ---");
System.out.println("Master correction: " + avg + "s");

for (int i = 0; i < n; i++) {
    int correction = avg - diff[i];
    System.out.println("Node " + (i + 1) + " correction: " + correction + "s");
```

```java
        }

        System.out.println("\n--- Synchronized Clocks ---");
        System.out.println("Master --> " + toHMS(masterTime + avg));

        for (int i = 0; i < n; i++) {
            int correction = avg - diff[i];
            System.out.println("Node " + (i + 1) + " --> " + toHMS(nodeTime[i] + correction));
        }

        sc.close();
    }
}
```

Output –

```
javac Berkeley.java
java Berkeley
Master time: 20:8:4
Enter number of nodes: 3
Enter time for Node 1 (h m s):
20 7 4
Enter time for Node 2 (h m s):
20 8 20
Enter time for Node 3 (h m s):
20 8 30

--- Time Differences ---
Node 1 difference: -60s
Node 2 difference: 16s
Node 3 difference: 26s
```

Average correction: -4s

--- Corrections ---

Master correction: -4s

Node 1 correction: 56s

Node 2 correction: -20s

Node 3 correction: -30s

--- Synchronized Clocks ---

Master --> 20:8:0

Node 1 --> 20:8:0

Node 2 --> 20:8:0

Node 3 --> 20:8:0